Canadian Monetary Policy and Unemployment

Evidence from Ontario, Quebec, Alberta and British Columbia | April 2009 - September 2026

Summary

This project looks at the relationship between the Bank of Canada's target overnight interest rate and unemployment in four provinces. I compared five regression specifications using monthly observations. Higher interest rates were associated with lower unemployment in the models using current, three-month and six-month rates. The estimated relationship was no longer statistically significant after year controls were included. These results describe patterns in the data; they do not show that higher interest rates cause unemployment to fall.

Research question

How does the relationship between the policy interest rate and unemployment change when I use different time lags and controls?

Data

I used the Bank of Canada's V39079 overnight-rate series and seasonally adjusted unemployment rates from Statistics Canada Table 14-10-0287-01. The unemployment variable is the simple average of the rates for Ontario, Quebec, Alberta and British Columbia. Each province receives the same weight, so this is **not** Canada's official unemployment rate. The data cover 210 months from April 2009 to September 2026 in the model without lags.

Method

I estimated ordinary least squares regressions using the current policy rate and rates lagged by three, six and twelve months. Each specification includes month-of-year indicators. I also estimated a three-month lag model with year indicators to check whether the result changes when common annual conditions are taken into account. Standard errors use the Newey-West method with twelve lags to allow for autocorrelation and heteroskedasticity.

Regression results

Model	Coefficient	HAC SE	p-value	Months
Current rate	-0.481	0.144	0.0008	210
3-month lag	-0.344	0.115	0.0027	207
6-month lag	-0.213	0.099	0.0312	204
12-month lag	-0.003	0.097	0.9781	198
3-month lag + year indicators	-0.198	0.222	0.3729	207

Coefficients are percentage-point differences in the unemployment measure associated with a one-percentage-point higher policy rate, conditional on the controls in each model. The models use different numbers of months.

Unemployment across the four provinces

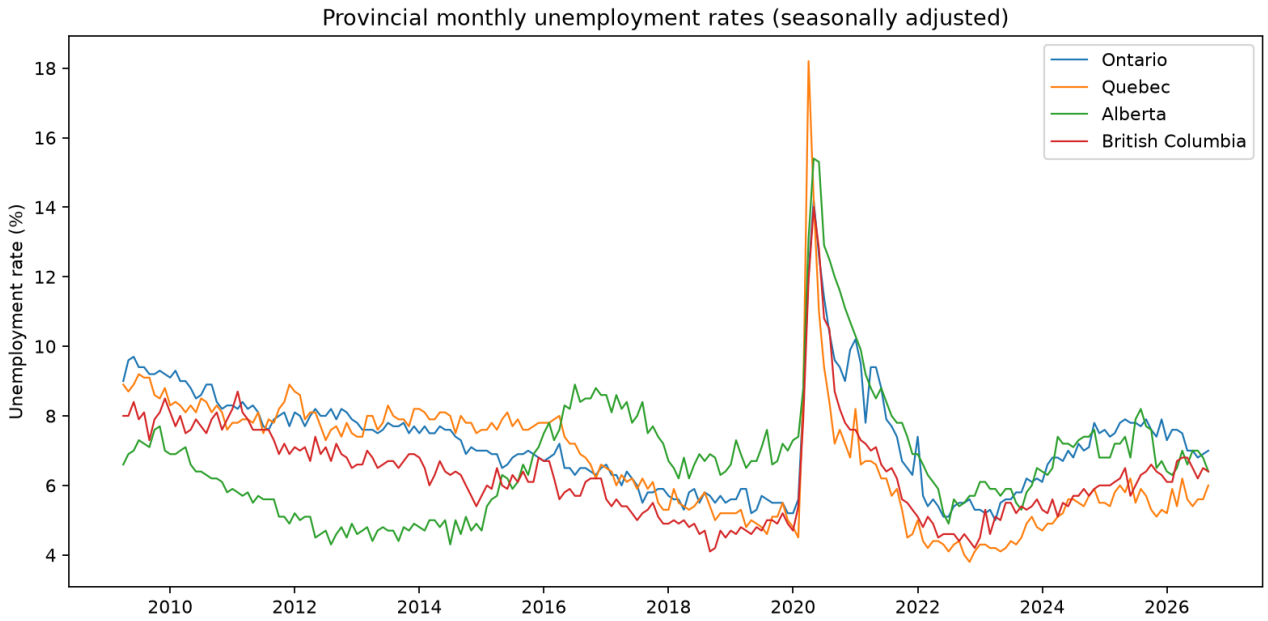


Figure 1. Seasonally adjusted provincial unemployment rates. The jump in 2020 is visible in all four provinces.

Bank of Canada overnight rate

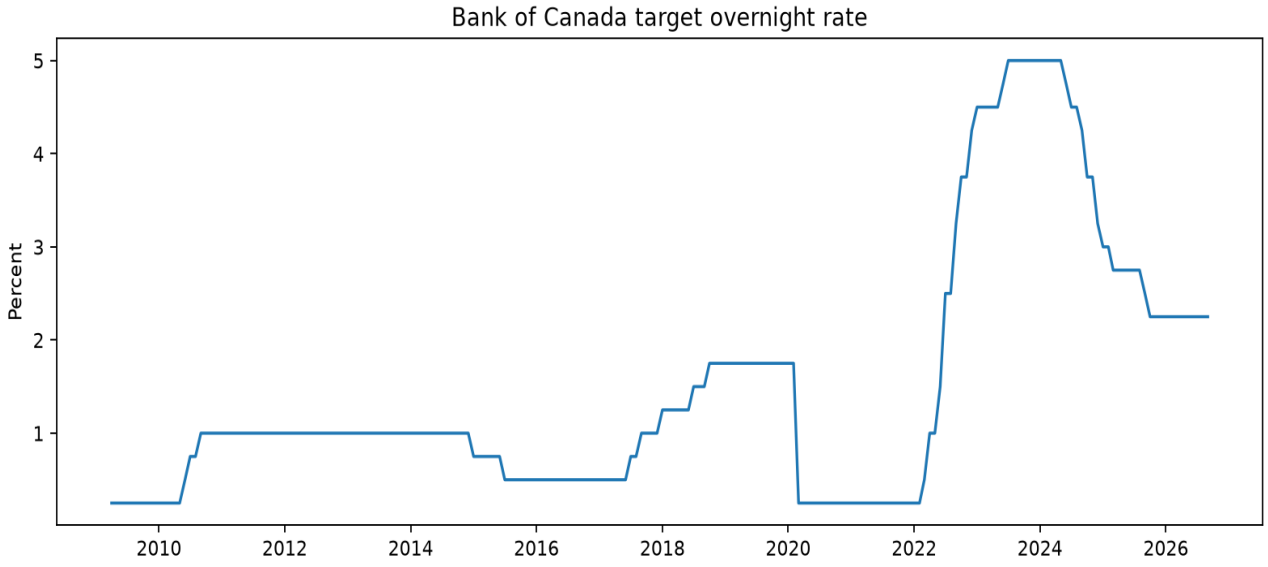


Figure 2. Monthly target overnight interest rate. The rate fell sharply in 2020 and rose from 2022.

Discussion

The current-rate, three-month and six-month models have negative coefficients with p-values below 0.05. The twelve-month estimate is close to zero. When I add year indicators to the three-month regression, its coefficient changes from -0.344 to -0.198 and the p-value increases from 0.0027 to 0.3729. This suggests that the result depends on the specification. Because lagged models also drop different numbers of early observations, the coefficients are not fully comparable.

The negative estimates should not be interpreted as evidence that interest-rate increases reduce unemployment. The Bank of Canada adjusts rates in response to economic conditions, and stronger economic activity can coincide with both higher rates and lower unemployment. The pandemic, inflation, fiscal policy and other shocks may also influence the observed relationship. Including year indicators accounts for some common conditions, but also removes much of the variation in policy rates from one year to another.

Limitations

This is a descriptive analysis. It covers four provinces and uses an equally weighted average rather than the national unemployment rate. The regressions do not fully account for inflation, output, previous unemployment or structural changes. They also do not isolate unexpected monetary-policy decisions. For these reasons, the coefficients cannot be used to estimate the causal effect or timing of monetary policy on unemployment.

Conclusion

The main finding is that the observed relationship between interest rates and unemployment changes when different lags and controls are used. The early specifications show a negative association, but the three-month relationship is not statistically significant after adding year indicators. The exercise shows why regression results need to be checked against alternative specifications before drawing economic conclusions.

Data and code

The accompanying Python script downloads public data, selects seasonally adjusted observations, creates monthly lag variables and estimates the regressions. The repository contains the code, figures and comparison table. The figures and estimates in this report were produced in a local run on October 10, 2026.

Bank of Canada: Valet API, series V39079, bankofcanada.ca/valet/observations/V39079/json
Statistics Canada: Table 14-10-0287-01, statcan.gc.ca/t1/tbl1/en/tv.action?pid=1410028701